Dr. Tim Hallatt

ASTROPHYSICIST

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Academic Positions

MIT Kavli Institute for Astrophysics and Space Research *Cambridge, Massachusetts*

POSTDOCTORAL ASSOCIATE *Sept. 2024 - present*

- advisor: Prof. Sarah Millholland

Education

McGill University *Montréal, Quebec*

PHD, PHYSICS *Sept. 2021 - July, 2024*

- advisor: Prof. Eve J. Lee
- thesis title: "On the Formation of Planets in the Milky Way's Thick Disk"
- topic: theoretical astrophysics
- tools: `MESA` hydrodynamics/interior structure code, `REBOUND` dynamics code, `Python`, `Fortran`
- additional skills: machine learning with `scikit-learn`

McGill University *Montréal, Quebec*

MSC, PHYSICS *Sept. 2019 - Sept. 2021*

- advisor: Prof. Eve J. Lee
- thesis title: "Leveraging Exoplanet Occurrence Rates to Test Planet Formation Theory"
- topic: theoretical astrophysics

University of Western Ontario *London, Ontario*

HON. BSC, PHYSICS *Sept. 2015 - April, 2019*

- honours thesis advisor: Prof. Paul Wiegert
- thesis title:"The Dynamics of Interstellar Asteroids and Comets Within the Galaxy"
- topic: astrophysical dynamics

Publications

* - SUPERVISED STUDENT

PUBLISHED (FIRST AUTHOR)

[7] **Hallatt, T.**, Owen, J. E., Millholland, S. 2026. Revealing the Origin of Desert Dwellers via Stellar Obliquities. Astrophysical Journal Letters, vol. 1005, no. 1.

[6] **Hallatt, T.**, Millholland, S., 2026. Shedding Light on Desert Dwellers. Astrophysical Journal, vol. 997, no. 2.

[5] **Hallatt, T.**, Millholland, S., 2026. Coupled Planetary Interior and Tidal Evolution. Astrophysical Journal, vol. 997, no. 2.

[4] **Hallatt, T.**, Lee, E. J., 2025. On the Formation of Planets in the Milky Way's Thick Disk. Astrophysical Journal, vol. 979, no. 120.

[3] **Hallatt, T.**, Lee, E. J., 2022. Sculpting the sub-Saturn Occurrence Rate via Atmospheric Mass Loss. Astrophysical Journal, vol. 924, no. 9.

[2] **Hallatt, T.**, Lee, E. J., 2020. Can Large-Scale Migration Explain the Giant Planet Occurrence Rate? Astrophysical Journal, vol. 904, no. 2.

[1] **Hallatt, T.**, Wiegert, P., 2020. The Dynamics of Interstellar Asteroids and Comets within the Galaxy: an Assessment of Local Candidate Source Regions for 1I/'Oumuamua and 2I/Borisov. Astronomical Journal, vol. 159, no. 4.

SUBMITTED (FIRST AUTHOR)

Hallatt, T., Barker, A. J., Millholland, S. 2026. The Neptunian Final Parsec Problem: How Inertial Waves Enable Neptunes and Gas Giants to Survive Tidal Migration. Submitted to *ApJ*.

Liveoak, D.*, **Hallatt, T.**, Millholland, S. 2026. Shaving the Outskirts of Planetary Systems in the Galactic Bulge via Stellar Flybys. Submitted to *ApJL*; co-first author with student.

Published (Student-Lead)

[3] Sethi, R.*, Millholland, S., **Hallatt, T.**, 2026. Coupled Orbital and Interior Evolution of Sub-Neptunes. In press, Astrophysical Journal.

[2] Liveoak, D.*, **Hallatt, T.**, Millholland, S. 2026. Stability of Multiplanet Systems Through Hot Jupiter Destruction. Astrophysical Journal Letters, vol. 1002, no. 1.

[1] Mireles, I., Ulmer-Moll, S., Liveoak, D.* (incl. **Hallatt, T.**) 2026. Uncovering the Rapidly Evolving Orbits of the Dynamic TOI-201 System. Science Advances, vol. 12, issue 16.

Published (Nth-author)

[1] Cadieux, C., Plotnykov, M., Doyon, R., et al. (incl. **Hallatt, T.**), 2023. New Mass and Radius Constraints on the LHS 1140 Planets – LHS 1140 b is Either a Temperate Mini-Neptune or a Water World. Astrophysical Journal Letters, vol. 960, no. 1.

Submitted (Nth-author)

Cui, K., Armstrong, D. et al. (incl. **Hallatt, T.**), 2026. Demographics of Close-In TESS Exoplanets Orbiting FGK Main-sequence Stars – II. Spectral-Type Dependence and the Neptunian Landscape (submitted to *MNRAS*).

King, G., Osborn, A., **Hallatt, T.**, 2026. Carving the Neptunian Desert as a Function of Stellar Mass (submitted to *MNRAS*).

White Papers

Benneke, B., Cowan, N., Rowe, J. et al. (incl. **Hallatt, T.**), 2019. Exoplanet instrumentation in the 2020s: Canada's pathway towards searching for life on potentially Earth-like exoplanets. Canadian Long Range Plan for Astronomy and Astrophysics White Papers, LRP2020. [link to paper]

Seminars & Presentations

December 2026. *From Tides to Flybys: Sculpting Exoplanetary Diversity Across the Galaxy*. Astrophysics Seminar, Canadian Institute for Theoretical Astrophysics, Toronto, Ontario (**invited**).

November 2026. *From Tides to Flybys: Sculpting Exoplanetary Diversity Across the Galaxy*. Astrophysics Seminar, Université de Montréal, Montréal, Quebec (**invited**).

November 2026. *From Tides to Flybys: Sculpting Exoplanetary Diversity Across the Galaxy*. Astrophysics Seminar, Trottier Space Institute, Montréal, Quebec (**invited**).

October 2026. *From Tides to Flybys: Sculpting Exoplanetary Diversity Across the Galaxy*. Astrophysics Seminar, UCLA, Los Angeles, California (**invited**).

October 2026. [Oral] *Sculpting Planetary Systems Probed by Roman via Stellar Flybys*. Lightning Talk, Planets in the Galactic Context, Center for Computational Astrophysics, New York.

September 2026. [Oral] *Revealing the Origin of Desert Dwellers via Stellar Obliquities*. The Formation, Life and Fate of Exo-Neptunian Systems, Lorentz Center, Leiden, Netherlands (**invited**).

June 2026. [Poster] *Revealing the Origin of Desert Dwellers via Stellar Obliquities*. Exoplanets 6 conference, Porto, Portugal.

June 2026. [Oral] *Revealing the Origin of Desert Dwellers via Stellar Obliquities*. AAS Division of Dynamical Astronomy meeting (DDA), Chicago, Illinois.

December 2025. [Oral] *Shedding Light on Desert Dwellers*. International Conference on Exoplanets and Planet Formation, Tsung Dao Lee Institute (TDLI), Shanghai, China (**invited**).

November 2025. *Exoplanet Demographics: A Journey Through Space and Time*. Astrophysics Colloquium, Southwest Research Institute (SwRI), Boulder, Colorado (**invited**).

October 2025. *Shedding Light on Desert Dwellers*. Exoplanet Lunch, Princeton University, Princeton, USA (**invited**).

June 2025. *Exoplanet Demographics: A Journey Through Space and Time*. Astrophysics Seminar, University of Bern, Bern, Switzerland (**invited**).

June 2025. *Exoplanet Demographics: A Journey Through Space and Time*. Astrophysics Seminar, Max Planck Institute for Astronomy, Heidelberg, Germany.

June 2025. *Exoplanet Demographics: A Journey Through Space and Time*. Astrophysics Seminar, Institute of Astronomy, University of Cambridge, Cambridge, UK (**invited**).

May 2025. *Shedding Light on Desert Dwellers*. [Oral] AAS Division of Dynamical Astronomy meeting (DDA), Atlanta, Georgia.

April 2025. *Shedding Light on Desert Dwellers*. Exoplanet Lunch, Harvard & Smithsonian Center for Astrophysics, Cambridge, USA.

October 2024. *On the Formation of Planets in the Milky Way's Thick Disk*. Planet & Star Formation Coffee, Max Planck Institute for Astronomy, Heidelberg, Germany (online; **invited**)

September 2024. *On the Formation of Planets in the Milky Way's Thick Disk*. [Oral] MIT & Harvard Planetary Meeting, Cambridge, USA.

June 2024. *On the Formation of Planets in the Milky Way's Thick Disk*. [Poster] Exoplanets 5 conference, Leiden, Netherlands.

September 2023. *Constraining Planet Formation Theory via Exoplanet Occurrence Rates*. Stars & Planets Seminar, Yale University, New Haven, USA (**invited**).

July 2023. *On the Formation of Planets in the Milky Way's Thick Disk*. [Oral] Towards Other Earths III: the Planet-Star Connection, Insitituto de Astrofísica e Ciências do Espaço, Porto, Portugal

June 2023. *On the Formation of Planets in the Milky Way's Thick Disk*. [Oral] Emerging Researchers in Exoplanet Science, Yale University, New Haven, USA.

May 2021. *Sculpting the sub-Saturn Occurrence Rate via Atmospheric Mass Loss*. [Oral] High Energy Exoplanets, European Space Agency XMM-Newton Workshop, Online.

November 2020. *Can Large-Scale Migration Explain the Giant Planet Occurrence Rate?* [Oral] ExoDem Conference, Caltech, Online.

October 2020. *Can Large-Scale Migration Explain the Giant Planet Occurrence Rate?* Exocoffee, Max Planck Institute for Astronomy (online; **invited**).

August 2020. *The Dynamics of Interstellar Asteroids and Comets Within the Galaxy*. [Oral] Division of Dynamical Astronomers Meeting, Online.

June 2020. *The Dynamics of Interstellar Asteroids and Comets Within the Galaxy*. [Poster] American Astronomical Society meeting, Online.

Mentorship

2025/2026	**DJ Liveoak**, Undergraduate/Graduate; primary research mentor for three projects on dynamics of TOI-201 system (Mireles et al. '26), hot Jupiters with companion planets (Liveoak et al. '26 a), and stellar flybys in the Galactic Bulge (Liveoak et al. '26 b)	*MIT*
2025/2026	**Ritika Sethi**, PhD student; research mentorship on coupled orbital/structural evolution of low-mass planets (Sethi et al. '26), and hot Jupiters in the Galactic Bulge (Sethi et al. in prep.)	*MIT*
fall, 2025	**Haedam Im**, Undergraduate; research mentorship on stellar rotation periods of hot Neptune hosts	*MIT*
summer, 2023	**Vincent Savignac**, Undergraduate; research mentorship on sub-Neptune core-envelope interaction	*McGill University*
2020-2021	**Didar Seghi**, Undergraduate; academic non-research mentorship	*McGill University*
2019-2020	**Griffin Schwartz**, Undergraduate; academic non-research mentorship	*McGill University*
2019-2020	**Harper Sewalls**, Undergraduate; academic non-research mentorship	*McGill University*

Media Citations & Interviews

Science News (2025): Galactic chaos at cosmic noon may have stunted Milky Way planet formation

Astronomy Magazine (2021): Our Galaxy's Marvelous Rogues and Misfits

Scientific American (2020): Mystery of Interstellar Visitor 'Oumuamua Gets Trickier

Nature (2019): How Two Intruders From Interstellar Space are Upending Astronomy

Populär Astronomi (2019): Interstellar comet Borisov is a well-known stranger

Select Awards & Fellowships

2021	**Alexander Graham Bell Canada Graduate Scholarship-Doctoral**, NSERC	*$ 105,000*
2021	**Perseverance Scholarship**, McGill University	*$ 1200*
2021	**L. Trottier Science Accelerator fellowship**, McGill University	*$ 5000*
2020	**Alexander Graham Bell Canada Graduate Scholarship-Master's**, NSERC	*$ 17,500*
2020	**Technologies for Exoplanetary Science Fellowship**, NSERC	*$ 6500*
2019	**Donald R. Hay Prize (for best thesis)**, Physics & Astronomy Dept., University of Western Ontario	*$ 300*
2019	**Dr. Gérard Hébert Scholarship in Physics (for community service, academic excellence, research potential)**, Physics & Astronomy Dept., University of Western Ontario	*$ 1700*

Service & Outreach

2026	**Participant**, MIT Leadership and Professional Strategies and Skills (LEAPS) Program	*MIT*
2020-present	**Peer reviewer**, AAS Journals, Monthly Notices of the Royal Astronomical Society, Astronomy & Astrophysics	
2026	**Postdoc Lunch coordinator**, MIT Kavli Institute	*MIT*
September, 2025	**Postdoc representative**, MIT Physics visiting committee	*MIT*
August, 2025	**Panelist/judge**, The Center for Excellence in Education's *Research Science Institute (RSI) encore presentations* (high school research program)	*MIT*
August, 2023	**McGill STEM summer camp**, Science Discussion/Q+A Group Leader	*McGill University*
2023	**Trottier Space Institute**, `arXiv` paper discussion organizer/leader	*McGill University*
2020-2022	**Trottier Space Institute**, meeting with seminar speaker organizer/leader	*McGill University*
2021-2022	**McGill Graduate Association of Physics Students**, VP Academic	*McGill University*
2021-2022	**McGill Graduate Association of Physics Students mentorship program**, lead organizer	*McGill University*
2019-2022	**McGill Graduate Association of Physics Students mentorship program**, mentor	*McGill University*
2021-2022	**McGill Graduate Association of Physics Students**, meeting with colloquium speaker organizer/leader	*McGill University*
2019-2022	**McGill Hackathon**, mentor	*McGill University*
2022	**Vanderbilt Astronomy Club, public lecture**, online	*Vanderbilt University*
2021	**AstroMcGill public lecture. Our Galactic Neighbourhood: Insights From Exoplanets and Interstellar Objects**, online	*McGill University*

Software

Author of `PSAND`: Planet Structure ANd Dynamics code package (published in Hallatt & Millholland 2025)